

Medvale Internal Medicine Chapter 13C

Chapter 13C

Appendix: Evidence-Based Medicine, Statistics, Consent, and Final Diagnostic Skills

Medvale Internal Medicine - final capstone segment for diagnostic reasoning, study interpretation, patient-centered decisions, and Graph-RAG extraction.

Chapter purpose

This appendix section turns the final pages of the book into a practical decision-making toolkit. The previous appendix segment focused on radiographs, ECGs, lines, nutrition, and antibiotic support skills. This segment focuses on how to judge evidence, how to interpret diagnostic tests, how to talk through decisions with patients, and how to avoid common clinical reasoning traps. The goal is not to make the reader a statistician; it is to make the reader safer when reading a study, choosing a test, weighing treatment harms, explaining uncertainty, or recognizing when a clean-looking algorithm is being applied to the wrong patient.

The uploaded appendix source emphasizes bedside interpretation and procedural support skills, including chest radiograph quality, chest and abdominal film patterns, ECG rate/rhythm/axis/interval interpretation, IV and central line complications, nutritional support, and antibiotic mechanisms. This chapter complements those skills by adding the cognitive and evidence-based framework that should sit behind every order and interpretation.

High-yield principle: diagnostic skill is not just recognizing the abnormal result. It is deciding whether the result should have been obtained, whether it changes probability enough to act, whether it is reliable in this patient, and whether the next step improves outcomes rather than merely producing more data.

1. The clinical question: converting uncertainty into a testable problem

Evidence-based medicine begins before the literature search. The clinician must first define the actual question being asked. A vague question such as, "Should this patient be anticoagulated?" contains multiple hidden questions: What is the diagnosis? What is the patient-specific baseline risk of thrombosis? What is the patient-specific bleeding risk? What intervention is being considered? What outcome matters most? What time horizon matters? What alternatives exist? A poorly framed question produces a poorly targeted search and often leads to overgeneralization from studies that do not match the patient.

PICO is a practical structure: Patient or Problem, Intervention or exposure, Comparison, and Outcome. For diagnostic questions, a modified structure is often more useful: patient population, index test, reference standard, outcome of interest, and intended management

action. A diagnostic test should not be ordered because it is available; it should be ordered because the result will move the patient from one management zone to another.

Question type	Clinical wording	Evidence focus	Common mistake
Therapy	In adults with unprovoked proximal DVT, does indefinite anticoagulation reduce recurrent VTE compared with 3-6 months of therapy?	Randomized trials, meta-analysis, absolute risk reduction, bleeding harms	Quoting relative risk reduction without baseline risk or bleeding risk.
Diagnosis	In an older adult with low pretest probability PE, does a negative D-dimer safely avoid CT pulmonary angiography?	Sensitivity, specificity, likelihood ratios, disease prevalence, rule-out threshold	Ordering the test in a patient whose pretest probability is too high or too low for the result to help.
Prognosis	After hospitalization for decompensated heart failure, what is the risk of readmission or death within 30-90 days?	Cohort studies, risk models, calibration, competing risks	Assuming a risk score applies outside its derivation population.
Harm	What is the risk of GI bleeding after NSAID initiation in an older adult taking anticoagulation?	Observational studies, pharmacovigilance, biologic plausibility, confounding control	Dismissing harm because randomized trials were underpowered to detect it.
Screening	Should an asymptomatic adult undergo a screening test for disease X?	Disease burden, test accuracy, early-treatment benefit, false positives, overdiagnosis	Assuming earlier detection is always beneficial.

2. Diagnostic testing: sensitivity, specificity, and likelihood ratios

Sensitivity and specificity describe a test's behavior in people whose disease status is already known. Sensitivity is the proportion of diseased patients with a positive test. Specificity is the proportion of nondiseased patients with a negative test. These are useful test properties, but they do not directly answer the patient's question: "What is the chance I have the disease given this result?" For that question, pretest probability and predictive values matter.

A highly sensitive test is useful for ruling out disease when negative if the test is applied in the right clinical setting. A highly specific test is useful for ruling in disease when positive if false

positives are uncommon and the result fits the clinical picture. The memory aids SNOUT and SPIN are useful but incomplete: Sensitive Negative rules OUT, Specific Positive rules IN. They fail when pretest probability is extreme, sampling is poor, timing is wrong, or the disease spectrum differs from the study population.

Measure	Formula	Clinical meaning	Pitfall
Sensitivity	$TP / (TP + FN)$	Among patients with disease, how often is the test positive? High sensitivity helps rule out when negative.	May be lower in early disease, mild disease, vaccinated hosts, partial treatment, or poor sampling.
Specificity	$TN / (TN + FP)$	Among patients without disease, how often is the test negative? High specificity helps rule in when positive.	May fall when related conditions cross-react or when disease definitions are broad.
Positive predictive value	$TP / (TP + FP)$	Probability of disease after a positive test in this tested population.	Changes dramatically with prevalence/pretest probability.
Negative predictive value	$TN / (TN + FN)$	Probability of no disease after a negative test in this tested population.	High in low-prevalence settings even for mediocre tests.
Positive likelihood ratio	$Sensitivity / (1 - specificity)$	How much a positive result increases odds of disease. $LR+ > 10$ is often strong.	Do not apply if the patient differs greatly from study population.
Negative likelihood ratio	$(1 - sensitivity) / specificity$	How much a negative result decreases odds of disease. $LR- < 0.1$ is often strong.	A negative test may not rule out disease when pretest probability is very high.

Pretest probability and post-test probability

Pretest probability is the clinician's estimated probability of disease before a new test result is known. It can come from a validated score, a population prevalence estimate, or a gestalt estimate grounded in symptoms, risk factors, and examination findings. Testing works best in the middle range of uncertainty. If pretest probability is already very low, a positive test may be false-positive noise. If pretest probability is already very high, a negative test may not safely exclude disease.

Likelihood ratios convert pretest odds to post-test odds. Odds are probability divided by one minus probability. Post-test odds equal pretest odds multiplied by the likelihood ratio. Post-test probability equals odds divided by one plus odds. In practice, clinicians often use rough probability zones rather than exact math: below the test threshold, do not test; between test and treatment thresholds, test; above the treatment threshold, treat or obtain confirmatory testing only if it changes risk, route, or urgency.

Example: if PE pretest probability is 5% and a test has LR- 0.1, post-test probability falls to roughly 0.5%, often below a rule-out threshold. If pretest probability is 70%, the same LR- leaves a post-test probability near 19%, which may still be too high to ignore. The same test result has different meaning in different patients.

Diagnostic thresholds

Zone	Probability range concept	Clinical behavior	Example
Below testing threshold	Disease probability is so low that test harms exceed likely benefit.	Do not test; provide reassurance, return precautions, or monitor.	No PE risk factors, normal vitals, clear viral pleuritic chest discomfort; PERC-negative patient.
Testing zone	Uncertainty is high enough that a result can change management.	Use the safest sufficiently accurate test first; choose test based on rule-in/rule-out goal.	Intermediate probability appendicitis; ultrasound or CT depending age/sex/pregnancy/resources.
Above treatment threshold	Disease probability is high enough that treatment or urgent action should begin.	Treat while confirming when delay is dangerous or confirmation changes management.	Suspected bacterial meningitis: antibiotics should not wait for perfect diagnostic certainty.

3. Study design and hierarchy of evidence

Study design determines what kinds of conclusions can be drawn. Randomized controlled trials reduce confounding when asking about therapy, but they may exclude older, sicker, pregnant, immunocompromised, or multimorbid patients. Cohort studies are often better for prognosis and harms. Case-control studies are efficient for rare outcomes but are vulnerable to recall and selection bias. Cross-sectional studies estimate prevalence and associations at a point in time but usually cannot establish temporal sequence. Case reports and case series are useful for signal detection, rare presentations, and hypothesis generation, not for estimating treatment effect.

Design	Best for	Strength	Main limitation
Randomized controlled trial	Therapy, prevention, some screening interventions	Randomization balances measured and unmeasured confounders when allocation and follow-up are rigorous.	May be expensive, short, underpowered for rare harms, and less generalizable.
Systematic review/meta-analysis	Summarizing multiple studies	Can increase precision and identify consistency across evidence.	Only as good as included studies; heterogeneity and publication bias can mislead.
Prospective cohort	Prognosis, incidence, exposures, harms	Temporal relationship is clear; can study multiple outcomes.	Confounding and loss to follow-up remain major threats.
Case-control	Rare outcomes or long latency diseases	Efficient when outcome is uncommon.	Selection bias and exposure measurement problems.
Cross-sectional	Prevalence, diagnostic associations	Fast snapshot of disease and risk factors.	Cannot usually determine cause precedes effect.
Case report/series	Rare disease, new adverse event, unusual presentation	Generates clinical awareness and hypotheses.	No denominator; cannot estimate risk or efficacy.

4. Measures of treatment effect

Treatment benefit should be interpreted in absolute as well as relative terms. Relative risk reduction can sound impressive even when the absolute benefit is small. Absolute risk reduction depends on baseline risk and is often more useful for counseling. Number needed to treat is the number of patients who must receive the intervention for a specified time to prevent one additional outcome. Number needed to harm is the analogous measure for adverse outcomes. Always attach a time horizon: an NNT over 5 years is not the same as an NNT over 30 days.

Measure	Formula or definition	Interpretation
Risk	Events / total patients	Probability of an outcome over a specified time.
Relative risk (RR)	Risk in treatment or exposure group / risk in control group	RR <1 suggests lower risk; RR >1 suggests higher risk. Needs absolute risk context.

Measure	Formula or definition	Interpretation
Odds ratio (OR)	Odds of outcome in group A / odds in group B	Approximates RR only when outcome is rare; can exaggerate apparent effect when outcome is common.
Hazard ratio (HR)	Relative event rate over time	Used in time-to-event analysis; assumes proportional hazards unless otherwise modeled.
Absolute risk reduction (ARR)	Control event rate - treatment event rate	Direct estimate of how much risk is lowered.
Number needed to treat (NNT)	1 / ARR	Patients treated for a defined time to prevent one event. Smaller NNT means larger absolute benefit.
Absolute risk increase (ARI)	Treatment harm rate - control harm rate	Direct estimate of added harm.
Number needed to harm (NNH)	1 / ARI	Patients exposed for a defined time to cause one additional harm.

Worked interpretation pattern

When reading a therapy trial, translate the abstract into a bedside sentence: “In patients like this, over this time period, this intervention changed this patient-important outcome by this absolute amount, with these harms, and the study was or was not powered to detect them.” Avoid phrases such as “significantly reduced” unless you can also describe the absolute effect and clinical relevance.

Trial result	Poor interpretation	Better clinical interpretation
RR 0.75 for MI, p=0.03	The drug reduced MI by 25%.	The relative risk was lower, but the clinical impact depends on baseline MI risk and absolute risk reduction.
ARR 1% over 5 years	The drug works.	Treating 100 similar patients for 5 years prevents about 1 event, before considering adverse effects, burden, and cost.
NNH 50 for major bleeding	Bleeding risk is acceptable.	For every 50 treated over the study period, about 1 additional major bleed occurred; compare directly with NNT for prevented

Trial result	Poor interpretation	Better clinical interpretation
Composite endpoint positive	All outcomes improved.	events. Check whether benefit is driven by soft endpoints, revascularization, or hospitalization rather than death or irreversible morbidity.

5. Confidence intervals, p-values, and clinical significance

A p-value answers a narrow question: if there were truly no difference between groups, how surprising would the observed data or more extreme data be? It does not measure effect size, clinical importance, probability the hypothesis is true, or probability the result will replicate. A confidence interval gives a range of effect sizes compatible with the data and model assumptions. Wide intervals indicate imprecision. A statistically significant but tiny effect may not matter clinically; a clinically important effect may fail to reach statistical significance if the study is underpowered.

Result pattern	Interpretation	Clinical response
Narrow CI entirely beneficial	Precise estimate favoring intervention.	Consider adoption if outcomes are patient-important and harms/costs acceptable.
Wide CI crossing no effect	Uncertain; study may be underpowered or heterogeneous.	Do not overstate absence of benefit; look for larger studies or meta-analysis.
Narrow CI around trivial effect	Effect is precise but likely not clinically meaningful.	Avoid changing practice solely because $p < 0.05$.
CI includes important harm and benefit	Evidence is too imprecise for confident decision-making.	Use patient values, alternatives, and risk tolerance; consider additional evidence.

6. Bias, confounding, and study validity

Internal validity asks whether the study's result is likely true for the studied patients. External validity asks whether the result applies to the patient in front of you. A beautifully conducted trial can be internally valid but externally weak for a frail older adult with CKD, polypharmacy, and multiple competing risks. Conversely, a messy observational dataset may be more representative but more vulnerable to confounding.

Threat	Definition	Clinical example	How to look for it
Selection bias	Groups differ because of how patients	Healthier patients are more likely to	Compare baseline characteristics;

Threat	Definition	Clinical example	How to look for it
	entered the study.	receive a procedure and appear to do better.	assess inclusion/exclusion and referral patterns.
Confounding	A third factor is associated with both exposure and outcome.	Coffee appears associated with lung cancer because smoking differs between groups.	Look for randomization, restriction, matching, stratification, regression, propensity methods.
Measurement bias	Exposure or outcome is measured differently or inaccurately.	Unblinded outcome assessors classify subjective improvement differently.	Check blinding, objective endpoints, standardized definitions.
Lead-time bias	Earlier diagnosis appears to prolong survival without changing death time.	Screening detects cancer earlier but mortality is unchanged.	Prefer disease-specific mortality and all-cause outcomes over survival from diagnosis.
Length-time bias	Screening preferentially detects slower-growing disease.	Indolent tumors are overrepresented in screen-detected cancers.	Look for randomized screening mortality trials and overdiagnosis estimates.
Publication bias	Positive studies are more likely to be published.	Small positive trials dominate the visible literature.	Look for trial registries, funnel plots, negative unpublished data.
Immortal time bias	A period during which outcome cannot occur is misclassified.	Patients must survive long enough to receive exposure, making exposure look protective.	Check time-zero alignment and time-varying exposure methods.

7. Screening and preventive testing

Screening is testing asymptomatic people to identify disease earlier than clinical presentation. Screening is beneficial only when the disease is important, the preclinical phase is detectable, the test is acceptable and accurate enough, early treatment improves patient-important outcomes, and harms are justified by benefits. Harms include false positives, anxiety, invasive follow-up, overdiagnosis, overtreatment, radiation, cost, and opportunity cost.

The most common screening error is equating earlier detection with better outcomes. Earlier detection can appear to improve 5-year survival even when mortality is unchanged because the disease clock started earlier. For screening, mortality reduction and quality-adjusted outcomes matter more than survival from diagnosis.

Screening question	What to ask before ordering
Who?	Does the patient match the age, risk, sex, pregnancy status, and life expectancy group for which screening is recommended?
What test?	Is this the recommended screening modality, or is it a diagnostic test being used inappropriately as screening?
How often?	What is the interval, and was the last result normal, abnormal, or incomplete?
What next?	If positive, is the patient willing and able to complete confirmatory testing and treatment?
When to stop?	Does limited life expectancy, high comorbidity burden, or patient preference make future benefit unlikely?

8. GRADE, guideline strength, and why recommendations differ

Guidelines differ because they weigh evidence, harms, costs, values, and feasibility differently. A strong recommendation usually means most informed patients would choose the recommended action and clinicians should generally offer it. A conditional or weak recommendation means decisions should be individualized, often because benefits and harms are closely balanced, evidence is uncertain, or patient values vary widely. Guidelines can also become outdated when new evidence, drug prices, resistance patterns, screening technology, or population risk changes.

GRADE concept	Meaning at the bedside
High certainty evidence	Further research is unlikely to change confidence substantially; still assess applicability to the patient.
Moderate certainty evidence	Further research may change the estimate or confidence; use while watching for updates.
Low/very low certainty evidence	Effect estimate is uncertain; individualization and shared decision-making become central.
Strong recommendation	Most patients should receive the intervention unless contraindicated or inconsistent with goals.
Conditional recommendation	Patient preferences, comorbidities, local

GRADE concept	Meaning at the bedside
	resources, and alternative strategies should drive the decision.

9. Diagnostic stewardship and overuse

Diagnostic stewardship means selecting tests that are likely to improve decisions while avoiding tests that create false positives, incidentalomas, downstream procedures, or treatment cascades. Overuse is not just wasteful; it can harm patients. Common examples include imaging uncomplicated acute low back pain without red flags, repeating stable labs without a decision point, ordering broad autoimmune panels without a compatible syndrome, testing for low-prevalence disease after a nonspecific symptom, and using tumor markers as vague screening tests.

A good test order contains an action plan: “If positive, I will do X; if negative, I will do Y; if indeterminate, I will do Z.” If no branch changes management, the test is probably not ready to be ordered.

Before ordering	Practical checkpoint
Clinical syndrome	Is there a coherent syndrome or risk factor pattern that makes the disease plausible?
Pretest probability	Is probability in the testing zone rather than extremely low or already high?
Test performance	Is the test accurate for this stage, population, specimen, and timing?
Downstream plan	What will change if the result is positive, negative, or equivocal?
Harms	Could the result trigger unnecessary imaging, biopsy, antibiotics, anticoagulation, anxiety, or labeling?
Patient values	Does the patient want the possible downstream pathway the test may initiate?

10. Capacity, informed consent, and shared decision-making

Capacity is decision-specific and time-specific. A patient may lack capacity for one complex decision while retaining capacity for a simpler one. Capacity is not the same as agreeing with the clinician. A capacitous patient can refuse a recommended therapy, even a life-sustaining one. The clinician’s task is to assess whether the patient can understand relevant information, appreciate the situation and consequences, reason about options, and communicate a stable choice.

Capacity element	What it means	Example bedside prompt
Understand	Can the patient describe the condition, proposed intervention, alternatives,	Can you tell me what you understand about what is happening and what we are

Capacity element	What it means	Example bedside prompt
Appreciate	and major risks/benefits? Can the patient apply the information to their own situation?	recommending? What do you think might happen to you if we do or do not proceed?
Reason	Can the patient compare options in a way that follows a recognizable logic?	What matters most to you as you weigh these choices?
Communicate choice	Can the patient express a consistent decision?	After talking through this, what would you like to do?

Informed consent is a communication process, not a signature. It should include the diagnosis or uncertainty, the proposed intervention, expected benefits, material risks, alternatives including no treatment, likely outcomes of refusing, and an opportunity for questions. Shared decision-making is especially important when multiple medically reasonable options exist or when benefits and harms depend heavily on patient preferences.

When the patient lacks capacity, the clinician should seek a legally authorized surrogate and use substituted judgment when possible: what would the patient have wanted based on prior statements, values, religion, goals, and life narrative? If unknown, decisions generally shift toward best interest: what best promotes the patient's welfare while minimizing burdens? Emergency treatment can proceed without formal consent when immediate action is needed to prevent serious harm and no surrogate is available.

11. Goals of care, code status, and end-of-life decision-making

Goals-of-care discussions translate medical options into patient-centered priorities. They should not start with a menu of procedures. First clarify understanding, prognosis, values, acceptable outcomes, fears, and tradeoffs. Then make a recommendation that connects medical choices to stated goals. Code status is only one part of goals of care; a patient can be DNR and still want hospitalization, antibiotics, transfusions, dialysis, or disease-directed therapy depending on the context.

Conversation step	Useful phrasing	Why it matters
Ask understanding	What have the doctors told you about where things stand?	Reveals misconceptions before recommendations.
Share prognosis	I am worried that time may be short, and I want to make sure we focus on what matters most to you.	Creates a realistic frame without abandoning care.
Explore values	What abilities or outcomes would be unacceptable to you?	Identifies thresholds for burdensome interventions.
Recommend	Because you said being	Converts values into a

Conversation step	Useful phrasing	Why it matters
	awake and comfortable with family is most important, I recommend we focus on comfort and avoid CPR if the heart stops.	medical plan.
Confirm plan	Let me make sure I have this right...	Prevents ambiguity and documents shared understanding.

12. Final integrated interpretation algorithms

Algorithm A: Should I order this test?

1. Define the clinical question in one sentence.
2. Estimate pretest probability using validated scores when available and clinical judgment when not.
3. Decide whether the patient is below the testing threshold, within the testing zone, or above the treatment threshold.
4. Choose a test whose strengths match the decision: sensitive rule-out test, specific rule-in test, staging test, safety test, or monitoring test.
5. Anticipate false positives, false negatives, incidental findings, and downstream actions.
6. Order only if at least one result branch changes management or counseling.

Algorithm B: How to interpret a positive test

1. Ask whether the test was indicated and whether specimen/timing/quality were appropriate.
2. Revisit pretest probability. A positive low-specificity test in a low-risk patient may be a false positive.
3. Look for confirmatory clinical features or a higher-specificity second test when the diagnosis carries major treatment harms.
4. Separate diagnosis from severity: a positive test may identify disease but not the need for admission, procedure, or lifelong therapy.
5. Communicate uncertainty explicitly and arrange follow-up for evolving findings.

Algorithm C: How to read a clinical trial abstract safely

1. Identify patient population and exclusions before reading the conclusion.
2. Determine whether the outcome is patient-important, surrogate, composite, or disease-oriented.
3. Extract absolute risks, not only relative effects.
4. Check confidence intervals, follow-up duration, loss to follow-up, and adverse events.
5. Ask whether the study patient resembles your patient in age, comorbidity, baseline risk, and competing risks.
6. Translate the result into a counseling sentence with NNT, NNH, time horizon, and uncertainty.

13. High-yield formulas and interpretation table

Concept	Formula	Use
Sensitivity	$TP / (TP + FN)$	Rule-out value when high and test is negative in appropriate population.
Specificity	$TN / (TN + FP)$	Rule-in value when high and test is positive in appropriate population.
PPV	$TP / (TP + FP)$	Probability of disease after positive result; increases with prevalence.
NPV	$TN / (TN + FN)$	Probability of no disease after negative result; increases when prevalence is low.
LR+	$Sensitivity / (1 - specificity)$	Multiplies pretest odds after a positive test.
LR-	$(1 - sensitivity) / specificity$	Multiplies pretest odds after a negative test.
Odds	$p / (1 - p)$	Needed for likelihood ratio math.
ARR	Control risk - treatment risk	Absolute benefit; basis for NNT.
NNT	$1 / ARR$	Number treated over defined time to prevent one outcome.
ARI	Treatment harm risk - control harm risk	Absolute harm; basis for NNH.
NNH	$1 / ARI$	Number exposed over defined time to cause one additional harm.
Relative risk	Risk exposed / risk unexposed	Association or treatment effect; needs absolute risk context.
Odds ratio	Odds exposed / odds unexposed	Common in case-control/logistic regression; approximates RR for rare outcomes.
Hazard ratio	Relative event rate over time	Time-to-event effect; check proportional hazards and absolute curves.

14. Final diagnostic red flags across the whole book

The safest clinician recognizes when a common complaint is no longer common. The following patterns should trigger escalation, urgent evaluation, or deliberate reconsideration of the differential diagnosis.

Presentation	Red flags	Do-not-miss diagnoses
Headache	Thunderclap onset, neurologic deficit, papilledema, immunosuppression/cancer, pregnancy/postpartum, age >50 with new headache, fever/meningismus	SAH, meningitis, cerebral venous sinus thrombosis, GCA, mass, intracranial hemorrhage
Chest pain	Hypotension, syncope, tearing back pain, pulse deficit, ischemic ECG, hypoxemia, pleuritic pain with PE risk	ACS, aortic dissection, PE, tension pneumothorax, tamponade, esophageal rupture
Dyspnea	Hypoxemia, altered mental status, silent chest, unilateral leg swelling, hemoptysis, fever/sepsis, new murmur	PE, pneumonia, COPD/asthma failure, heart failure, pneumothorax, ACS
Abdominal pain	Peritonitis, shock, pregnancy, GI bleeding, severe pain out of proportion, rigid abdomen, immunosuppression	AAA, ischemic bowel, perforation, ectopic pregnancy, appendicitis, obstruction
Back pain	Cancer, infection risk, fever, neurologic deficit, saddle anesthesia, urinary retention, trauma, chronic steroid use	Cord compression, cauda equina, epidural abscess, fracture, malignancy
Rash	Mucosal involvement, skin pain, fever, purpura, hypotension, necrosis, facial edema/eosinophilia, immunosuppression	SJS/TEN, meningococcemia, necrotizing infection, DRESS, TSS, vasculitis
Lab abnormality	K >6.5 or ECG changes, Na <120 or severe symptoms, severe acidosis, platelets very low with bleeding/schistocytes, lactate elevation with shock	Hyperkalemic arrhythmia, cerebral edema/osmotic injury, DKA/sepsis/toxin, TTP/DIC, shock states

15. Graph-RAG extraction map

For graph construction, this chapter creates nodes and edges that are methodologic rather than disease-specific. These nodes can connect across all prior organ-system chapters and improve retrieval when a user asks a symptom-first or decision-first question.

Node class	Example nodes	Useful edge types
Diagnostic-test concepts	sensitivity, specificity, likelihood ratio, predictive value, testing threshold	modifies probability of; rules out; rules in; depends on prevalence
Study-design concepts	RCT, cohort, case-control, meta-analysis, cross-sectional study	best evaluates; vulnerable to; supports recommendation
Effect-size concepts	ARR, NNT, NNH, relative risk, odds ratio, hazard ratio	quantifies benefit; quantifies harm; requires time horizon
Bias concepts	lead-time bias, length-time bias, confounding, selection bias, measurement bias	distorts estimate of; common in; mitigated by
Ethics concepts	capacity, informed consent, surrogate decision-maker, substituted judgment, best interest	required before; guides refusal of; used when lacks capacity
Clinical safety concepts	red flag, emergency threshold, diagnostic stewardship, incidentaloma	triggers escalation; argues against testing; causes downstream harm

References and source notes

Primary local source: uploaded appendix chapter pages emphasizing radiographic interpretation, ECG interpretation, IV therapy, central venous and arterial lines, nutrition support, parenteral nutrition, and antibiotic therapy. This chapter extends that appendix into evidence interpretation, diagnostic stewardship, consent, capacity, goals of care, and final diagnostic reasoning skills.

External references consulted for current framing: USPSTF methods and procedures for preventive recommendations; GRADE handbook and GRADE Working Group materials on certainty of evidence and strength of recommendations; AMA Code of Medical Ethics on informed consent and patient decision-making; ABIM Foundation Choosing Wisely materials on unnecessary tests and procedures.